

나노재료학(공) (MSE303-01)

학기	2026년 2학기
담당교수	이종수
이수구분	교선평수
학점	3.0
강의시간	화16:00-17:30(E7 - L21), 목16:00-17:30(E7 - L21)
강의실	
소속	기초학부

교과개요

Nanotechnology can create many new materials and devices in a wide range of applications. In modern nanoscience, various disciplines such as chemistry, physics, material engineering, device engineering, and analytical science are organically combined to lead the development of nanoscience. Therefore, in this lecture, we will study contents related to the synthesis and analysis of nanomaterials and device application from the viewpoints of nanomaterials and device related chemistry, physics, and material engineering in nanotechnology.

교과목표

양자점, 2차원 나노소재의 화학 및 재료관점 이해와 나노재료 분석법 및 이들 나노소재를 이용한 전자소자 및 광전 소자 응용을 재료적, 물리적인 이슈를 이해한다

주차별 강의계획

1주차:Overview of Nanotechnology

2주차:Basic of Crystal Structures

3주차:Basic of Optics

4주차:Optical Properties : Absorption Spectroscopy and Photoluminescence Spectroscopy

5주차:Structure Analysis : Electron Microscopy and Chemical Analysis

6주차:Quantum Dots or Artificial Atoms : Synthesis I

7주차:Quantum Dots or Artificial Atoms : Synthesis II

8주차:Mid-term exam

9주차:Quantum Dots or Artificial Atoms : Film Preparation and Charge Transport

10주차:Quantum Dots Application : Field-Effect Transistors

11주차:Quantum Dots Application : Photodetectors and Image sensors

12주차:Applications of Quantum Dots or Artificial Atoms : Photovoltaic Devices

13주차:Applications of Quantum Dots or Artificial Atoms : Display and Light Emitting Diodes

14주차:Two-Dimensional (2D) Materials and Applications : Graphen, Transition dichalcogenide Materials (TMDCs)

15주차:Nanodevice Fabrication and Characterization of Two-Dimensional (2D) Materials

16주차:Final Exam